

**Full Expensing and Firm Behaviors:
Evidence from the Hospital Industry and the Tax Cuts and Jobs Act of 2017**

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Abstract:

Full expensing allows firms to deduct the cost of investment, making it easier for firms to buy new equipment, and possibly leading to higher productivity, wages, and employment. This paper examines the effects of full expensing, which was a salient feature in the Tax Cuts and Jobs Act of 2017 (TCJA), on employment and investment within the hospital industry. TCJA reduced the corporate income tax rate and allowed full expensing of qualified machinery to stimulate economic growth and encourage business investment. The hospital sector comprises over 6.6 million jobs and is regarded for its vital societal and economic role. There are both nonprofit and for-profit hospitals, which provides an opportunity to consider the effects of full expensing since only for-profit hospitals were able to take advantage of the tax policy change. This study hypothesizes that TCJA had no significant impact on employment investment and labor compensation in for-profit hospitals relative to non-profit hospitals, owing to the full expensing policy and further based on economic theory. Annual hospital-level data from the Centers for Medicare & Medicaid Services from the period 2015 to 2022 were used for the analysis. The findings suggest that for-profit hospitals experienced lower labor employment and lower labor compensation than non-profit hospitals following TCJA. However, the result was significant only three years after the Act came into effect.

Keywords:

Tax Cuts and Jobs Act (TCJA), employment, tax policy, investment, hospitals, for-profit, non-profit, First Difference (FD), Difference in Differences (DID)

JEL Classifications:

H71: State and Local Taxation, Subsidies, and Revenue

J21: Labor Force and Employment, Size, and Structure

J31: Wage Level and Structure • Wage Differentials

K34: Tax Law

1. Introduction

The hospital industry has been economically vital due to its massive scale and essential societal responsibilities. Hospitals, among the largest employers in the U.S., provided over 6.6 million jobs, supported one in six jobs nationwide, and generated approximately 4.8 trillion dollars in total economic output in 2023 (American Hospital Association, 2025). Overall health care spending accounted for 17.6% of U.S. GDP, with hospital industry expenditure accounting for nearly a third of that (Martin et al, 2024). Such significance underscores the need to closely examine the effects of a major tax policy, like the Tax Cuts and Jobs Act of 2017 (TCJA), on this industry and the potential economic risks and benefits it entails. In particular, what would happen to the 6.6 million jobs and employment opportunities in this industry after the tax cuts?

The TCJA's corporate side provisions included a flat tax rate cut for top-earning corporations to 21% and allowed full expensing of equipment and machinery. TCJA replaced the graduated corporate tax rate structure with a flat 21% rate for corporations across all income levels, thereby lowering the effective tax rate for top-earning firms from 35% to 21% (IRS, 2018). This policy significantly increased the cash flow for for-profit hospitals relative to non-profit hospitals. Full expensing allowed firms to immediately deduct the cost of investment in qualified property, such as medical equipment or machinery, enabling more aggressive investing and hiring. These provisions directly benefited the taxable entities.

The hospital industry provides a unique setting for examining the effects of TCJA, which are seldom observed in other sectors of the economy. Most notably, the hospital industry comprises firms with various ownership structures, including private for-profit, private non-profit, and government hospitals. These hospitals participate and compete within the same market. This mix creates a natural environment for empirical analysis. To be more specific, given that this study focused on policies associated with tax cuts, a sector comprising taxable (for-profit hospitals) and tax-exempt entities (non-profit hospitals) is ideal, as all tax-exempt entities are naturally unaffected. Non-profit and for-profit hospitals differ in the services they provide; for instance, non-profit hospitals are generally larger and more likely to provide less profitable services, such as psychiatric emergency care. In contrast, for-profit hospitals are much more responsive to profitability and highly selective (Horwitz & Nichols, 2022). It is reasonable to assume that both types of hospitals share similar social and economic roles, as they provide essential care, employ large labor forces, and respond to local health needs. Under this assumption, this study treats non-profit and for-profit hospitals as functional comparables, satisfying the treatment and control group conditions for a difference in differences framework.

This paper hypothesizes that TCJA provisions affecting for-profit hospitals did not result in a significant change in labor investment and employment relative to non-profit hospitals. This proposal is supported and developed from a microeconomics theoretical model in which a firm maximizes profit subject to productivity constraint, implying limited flexibility and other priorities.

2. Literature Review

2.1 Impact of tax cuts on employment and wages

Tax cuts are a highly debated fiscal policy. Economists and politicians who support such a policy argue that on the firms' side, tax cuts reduce the burden on corporations, enabling them to expand and invest more aggressively, thereby generating more employment opportunities and potentially increasing wages. On the consumer side, economists argue that individual tax cuts increase disposable income, stimulate consumption, promote broader economic activity, and foster growth. Conversely, critics argue that tax cuts provide only short-term benefits while potentially causing long-term economic problems.

Ljungqvist & Smolyansky (2018) collected state-level tax data from 1969 to 2010 and studied the effects of tax cuts by comparing contiguous counties across state borders, which served as a control group. Finding that a 1% corporate tax cut leads to employment in the affected county rising by approximately 0.2% and total wage income rising by approximately 0.3%, measured relative to the neighboring county across the state. Ultimately, they reported that tax increases are generally harmful, while tax cuts appear to be effective only during recessions.

2.2 Hospital industry framework and TCJA

King (2019) analyzed the impact of the TCJA on the healthcare industry by examining CMS cost report data and secondary sources, such as IRS filing data, to ensure completeness. The main findings of this paper were that TCJA made capital investment more profitable and potentially led non-profit hospitals to cut less profitable but essential services. It also emphasized that policymakers failed to consider the unique competition-market dynamics in the hospital industry, which could lead to unintended consequences.

Fox & Pyle (2022) studied TCJA directly by examining the unique dynamics between banks and credit unions. Even though their study was not conducted in the hospital industry, they found a very similar setting: credit unions, which enjoy tax-exempt status, serve as close substitutes for small- to medium-sized banks. Thus, creating a treatment group and control group environment, and allowing for a DID framework. Fox & Pyle analyzed data on labor compensation, loan interest payments, and loan income from 2013 to 2019. Their study concluded that banks enjoyed about 40% of total tax savings, and there was no evidence that tax cuts benefited employees and customers.

Liew & Murphy (2024) examined readmission rates and financial records from California hospitals and analyzed investments to evaluate the effect of TCJA on the competitive dynamics of the hospital industry. Liew and Murphy concluded that the increase in cash flow for for-profit hospitals led to the rise in capital investment relative to non-profit hospitals. They also measured quality of care through readmission rates and found that, after TCJA, for-profit hospitals had lower readmission rates, suggesting an increase in medical care quality.

All three prior studies analyzed the economic impacts of TCJA, focusing on settings where non-profit and for-profit organizations compete in the same market. This paper builds on the framework implemented by

Liew & Murphy (2024). While their studies primarily focused on the effects of investment and quality of care in for-profit hospitals relative to their counterparts, this paper extends their framework by focusing more on employment and labor compensation data, examining labor investment and employment behavior in private for-profit hospitals.

3. Economics Theory and Model

3.1 Theory overview and background

The theory of a firm that maximizes profits subject to production constraints was formalized and developed by scholars such as Samuelson (1947), Hicks (1932), and Shepard (1953). Samuelson formalized optimization in economics using the theory of constrained optimization. The core idea of this theory is how profit-seeking firms choose the most efficient combination of resources, typically labor and capital, to produce a given output at the lowest possible cost. The optimization process also reveals that cost minimization is equivalent to profit maximization.

A few key assumptions are needed under such a framework. First, firms operate in a competitive input market, meaning they have no power over labor or capital prices. Second, firms' decisions are rational and based on economic efficiency. Third, firms' production affects only themselves, leaving no room for externalities. The real world is more complex, and sometimes these assumptions may not hold. However, economic models and theories provide a starting point for empirical research. This paper uses profit-maximization theory as a foundation for understanding and analyzing the profit-maximizing behavior of for-profit hospitals. Notably, the effects of said behaviors on employment and labor compensation in response to TCJA's lowered effective tax rates and allowing for capital investments' full expensing for for-profit hospitals compared to non-profit hospitals.

3.2 Model 1: Profit maximization before TCJA (higher corporate income tax rate and no full expensing)

This model assumes that firms use only fixed amounts of capital (K) and labor (L) as input resources and produce (Q) units of output using the production function (F):

$$Q = F(K, L) \quad [3.1]$$

In addition, this paper assumes that firms can not affect the market cost of capital (coc) and labor (w), and the functional form for the estimation of cost (C) is:

$$L \cdot w + K \cdot \text{coc} = C \quad [3.2]$$

The Hall-Jorgenson cost-of-capital equation is represented as coc in this model. coc operates with three additional variables: τ , δ , and $\hat{z}$. τ represents the corporate income tax rate, δ is the depreciation rate, and $\hat{z}$ is the present value of depreciation deduction that has a value between. $\hat{z}$ can also be interpreted as the incentives for firms to invest in capital.

The Hall-Jorgenson cost of capital (coc) formula:

$$coc = \frac{(r + \delta)(1 - \tau_z)}{(1 - \tau)} \quad [3.3]$$

To calculate the profit of a firm under this model, simply subtract the cost from total revenue, which is just the price of the product/service (P) times the quantity produced (Q):

$$\pi = P^*Q - wL - cocK \quad [3.4]$$

From the previous equation, Q can also be written as: $Q = F(K, L)$. Plug this back into the profit function:

$$\pi = P^*F(K, L) - wL - cocK \quad [3.5]$$

To simulate a firm's behavior, this model implements profit maximization after corporate income tax under production constraint. Because coc is a tax-adjusted rental price, it should not be double-counted as a separate deductible term inside the taxable-income bracket (Hall & Jorgenson, 1967).

The Lagrange Function form is denoted below as Z, and λ represents the Lagrange multiplier:

$$Z(K, L, \lambda) = (1 - \tau)(P^*F(K, L) - wL) - cocK + \lambda(Q - F(K, L)) \quad [3.6]$$

To solve the model, take the partial derivative of each variable: K, L, λ , and set each to zero to get the first-order condition. The first order conditions are listed below:

(The partial derivative of $F(K, L)$ with respect to K and L is written as F_K and F_L .)

$$\begin{aligned} \frac{\partial Z}{\partial K} &= (1 - \tau) * P^*F_K - coc - \lambda F_K = 0 \\ \frac{\partial Z}{\partial L} &= (1 - \tau) * P^*F_L - (1 - \tau) * w - \lambda F_L = 0 \\ \frac{\partial Z}{\partial \lambda} &= Q - F(K, L) = 0 \end{aligned}$$

Solving these three first-order conditions gives the following equation:

(The left-hand side of this equation is MRTS because F_L is the marginal product of labor, MPL, and F_K is the marginal product of capital MPK.)

$$\frac{F_L}{F_K} \text{ pre TCJA} = \frac{(1 - \tau)w}{coc} = \frac{(1 - \tau)w}{\frac{(r + \delta)(1 - \tau_z)}{(1 - \tau)}} \quad [3.6]$$

(Equation 1)

Equation 1 is the optimal bundle of labor and capital that a firm should use to maximize profit, subject to the production constraint. To examine the effect of TCJA, this study presents the second model, in which a firm faces a lower corporate income tax rate and full capital expensing.

3.3 Model 2: Profit maximization under corporate income tax and capital full expensing

Model 2 operates under the same conditions and assumptions as Model 1, except for a lower corporate income tax rate represented by τ' and $\hat{z} = 1$ to represent the full capital expensing provision

The cost of capital can be simplified as such when $\hat{z} = 1$:

$$coc = (r + \delta) \quad [3.7]$$

The optimization process with the Lagrange multiplier and first-order conditions:

Lagrange Formula:

$$Z(K, L, \lambda) = (1 - \tau') * [P * F(K, L) - wL] - cocK + \lambda (Q - F(K, L)) \quad [3.8]$$

First order conditions:

$$\frac{\partial Z}{\partial K} = (1 - \tau') * (P * F_K) - coc - \lambda F_K = 0$$

$$\frac{\partial Z}{\partial L} = (1 - \tau') * P * F_L - (1 - \tau')w - \lambda F_L = 0$$

$$\frac{\partial Z}{\partial \lambda} = Q - F(K, L) = 0$$

Solving for the optimal bundle:

$$\frac{F_L}{F_K} \text{ post TCJA} = \frac{(1 - \tau')w}{coc} = \frac{(1 - \tau')w}{r + \delta} \quad [3.9]$$

(equation 2)

$\frac{(1 - \tau')w}{r + \delta}$ (optimal bundle in model 2) is larger than $\frac{(1 - \tau)w}{(r + \delta)(1 - \tau\hat{z})}$ (optimal bundle in model 1) because $(1 - \tau')$ is

larger than $(1 - \tau)$ due to the lower tax rate after TCJA: τ' . In addition, because $0 < \hat{z} < 1$, $(r + \delta)$ is smaller than

$\frac{(r + \delta)(1 - \tau\hat{z})}{(1 - \tau)}$. Combined, the introduction of TCJA in this model disrupted the optimal bundle for firms, leaving

them operating in an unoptimized state. Under this model, firms can adjust only the quantities of labor and

capital they use. To return to the optimal situation, firms must increase F_L and decrease F_K . In other words, firms

will increase their marginal product of labor by hiring less labor and reduce their marginal product of capital by

using more capital. This theoretical model concludes that a tax-cut policy, such as TCA, with a lower corporate

income tax rate and full capital expensing, would incentivize private firms to invest more in capital and less in

labor. Such findings laid the foundation for this paper’s hypothesis that TCJA did not boost employment in for-profit hospitals.

4. Data

4.1 Data cleaning and variable selection

The research used Hospital Provider Cost Report Data from the Centers for Medicare & Medicaid Services (CMS). These datasets cover 2015-2022 and are publicly available online. After cleaning the primary dependent variables: FTE employees on payroll and total salaries, the dataset comprises 30923 observations of hospitals spanning 8 years. To ensure a suitable econometrics framework, all hospitals that changed their status from for-profit to non-profit, or vice versa, and hospitals that lacked complete records for the entire 8-year period were excluded. This research also treated government hospitals as non-profit for the difference in differences (DID) and first difference (FD) frameworks. Government hospitals are often exempt from federal and state corporate income taxes, as are private non-profit hospitals. In addition, government hospitals are publicly owned, and any surplus would be reinvested in their operations. Under these conditions, government hospitals closely resemble nonprofit private hospitals, given their employment behavior in response to tax-cut policies. Therefore, all government hospitals in the dataset were reclassified as private, non-profit hospitals. Consequently, the primary focus of this paper is to explore the impact of tax-cut policy on employment behavior across for-profit and non-profit hospitals. The DID and FD frameworks rely on a clear distinction between treatment and control groups, and hospitals that switched status over the eight years could violate this setting and the parallel trend assumptions. Removing these observations is necessary for unbiased and accurate regression estimation. Moreover, hospitals with incomplete data over the eight years may also bias model estimates, as the framework requires consistent data to measure changes within hospitals, and incomplete data could distort these estimates.

This study created a binary variable for the type of control, which is equal to 1 if the hospital is for-profit. This process reduces the number of observations to 21775 while ensuring data completeness. Table 1 summarizes the distribution of hospital types. This result is expected, according to statistics from the Department of Health and Human Services (2023), which indicate that about 36.1% of hospitals are for-profit, while the rest are either non-profit private hospitals (49.2%) or government hospitals (14.7%). The distribution in this dataset (Table 1) after cleaning is slightly different from that of real-world data, but it still exhibits sufficient variation and robustness to support statistical inference.

(Table 1) Distribution of Hospital Types		
For-profit	920	25.27%

Some states have more favorable legislation for non-profit hospitals, for example, New York State's restriction on publicly traded for-profit companies owning hospitals (New York Health Foundation, 2018). Due to these policies, some states have a limited number of for-profit hospitals. This disproportionate distribution of for-profit and non-profit hospitals can bias DID and FD regression estimates when there is no treatment group in the state. To counter this potential bias in the estimation, states with fewer than 5% for-profit hospitals were excluded. A 5% threshold is a reasonable choice because it ensures meaningful control and treatment groups while also preserving the number of observations and the data's variance. Table 2 lists the states excluded from the dataframe.

(Table 1) Distribution of Hospital Types		
Non-profit	2720	74.73%

(Table 2) States Excluded from DID Analysis

State	Percent For-Profit Hospitals
CT	0.0
HI	0.0
MD	0.0
ME	0.0
VI	0.0
VT	0.0
MN	0.9
NY	3.3
IA	3.5
MT	4.4
OR	4.5
WA	4.5

After removing all these states, the CMS provider cost dataframe has 17815 observations. The final cleaned dataframe contains 3563 hospitals across 5 years from 43 states. A substantially large dataset like this could provide sufficient variance and statistical robustness to identify the impacts of TCJA on the hospital industry.

The primary interest is to measure the impacts of TCJA on employment and labor compensation. To measure these metrics, this research used FTE Employees on Payroll and Total Salaries from the CMS dataset as the primary dependent variables to quantify the specific effect. The CMS collected data for FTE Employees on Payroll by adding up all the hours worked by full-time employees in the first week of each quarter, then dividing

by 160 for the full quarter and by 80 for the semi-quarter. This variable represents both the headcount and the hours worked for the labor force. It is a precise, quantifiable measure of the total labor force available to hospitals, making it helpful for this study to explore the main question. The other primary dependent variable is Total Salaries. CMS collected data on this variable from total salary expense recorded in the hospital's accounting books or balance sheet. While FTE Employees measured the quantity and size of the labor force, total salaries measured the overall labor compensation. These two variables complement each other, providing quantitative measures of the overall labor force and labor compensation in the hospital industry. The primary explanatory variables for this study are binary variables for for-profit status and the post-TCJA period. Some other minor variables are included as control variables or fixed effects, such as the number of beds, state code, total discharge, total cost, and a binary variable indicating whether the location is urban or rural. These minor variables either control for hospital size or control for state-level fixed effects.

4.2 Summary Statistics

This section focuses on the characteristic statistics of the cleaned data set—particularly the distributions and key statistics of variables from non-profit and for-profit hospitals. Characteristic statistics tables for the variables of interest are calculated and grouped by treatment (for-profit hospitals) and control (non-profit hospitals). Tables 3 and 4 present the summary statistics for all numeric variables, including two variables of primary interest: FTE Employees and Total Salaries for all the non-profit hospitals in the data. For most variables, non-profit hospitals have higher means and maximums, as expected, because they are typically larger and therefore have bigger labor forces, higher salaries, and greater costs.

(Table 3) Descriptive Statistics of Numeric Hospital Variables

Variable	Mean	SD	Min	Max	Median
FTE Employees	915.39	4,433.13	0.07	641,862.8	296.64
Total Salaries	70,318,958.46	149,211,890.79	67,757.00	3,114,335,709.0	20,939,222.00
Number of Beds	137.35	180.41	1.00	3,716.0	65.00
Total Discharges	1,600.16	2,467.40	1.00	22,164.0	478.00
Total Costs	156,851,763.49	296,363,514.07	989,453.00	4,524,701,331.0	46,318,918.50

(Table 4) Descriptive Statistics by Ownership Type

Ownership Type	Variable	Mean	SD	Min	Max	Median
Non-Profit	FTE Employees	1140.74	5256.99	0.07	641862.8	388.34
Non-Profit	Total Salaries	88447823.01	172813683.51	67757.00	3114335709.0	28642853.50
Non-Profit	Number of Beds	150.89	201.95	1.00	3716.0	59.00
Non-Profit	Total Discharges	1836.23	2744.84	1.00	22164.0	501.00
Non-Profit	Total Costs	194247378.44	340706088.36	989453.00	4524701331.0	59753760.50
For-Profit	FTE Employees	378.69	481.04	8.38	9346.0	199.83
For-Profit	Total Salaries	27142580.85	38950754.28	457257.00	597594403.0	12770071.00
For-Profit	Number of Beds	105.11	107.08	7.00	1498.0	72.00
For-Profit	Total Discharges	1037.94	1481.83	1.00	14090.0	449.00
For-Profit	Total Costs	67788977.01	96406372.29	1387022.00	996868239.0	29776025.00

4.3 Parallel Trend

Two models were employed for causal inference analysis: the DID estimation and the FD estimator. Both of these two estimations required data that satisfy the parallel trend assumption. Under these conditions, it is crucial to carefully examine the parallel trends of the primary dependent variables to perform causal inference analyses and ensure unbiased results. The natural logs of both variables were used for several reasons. First, hospitals vary in size, with some having substantially larger labor forces than others. For example, Cleveland Clinic Hospital had over 26000 FTE employees, while ST. Francis Hospital Wilmington had only about 886 FTE employees. These drastically different numbers could make the effect uneven and difficult to compare. Using the natural log in this case eliminated the significant difference, making the results more even and comparable. Second, the natural log smoothed the distribution of the data, especially by counteracting potential bias from outliers. Using the mean of the logged primary dependent variable and grouping by years from 2015 to 2022, this study plotted parallel trend graphs.

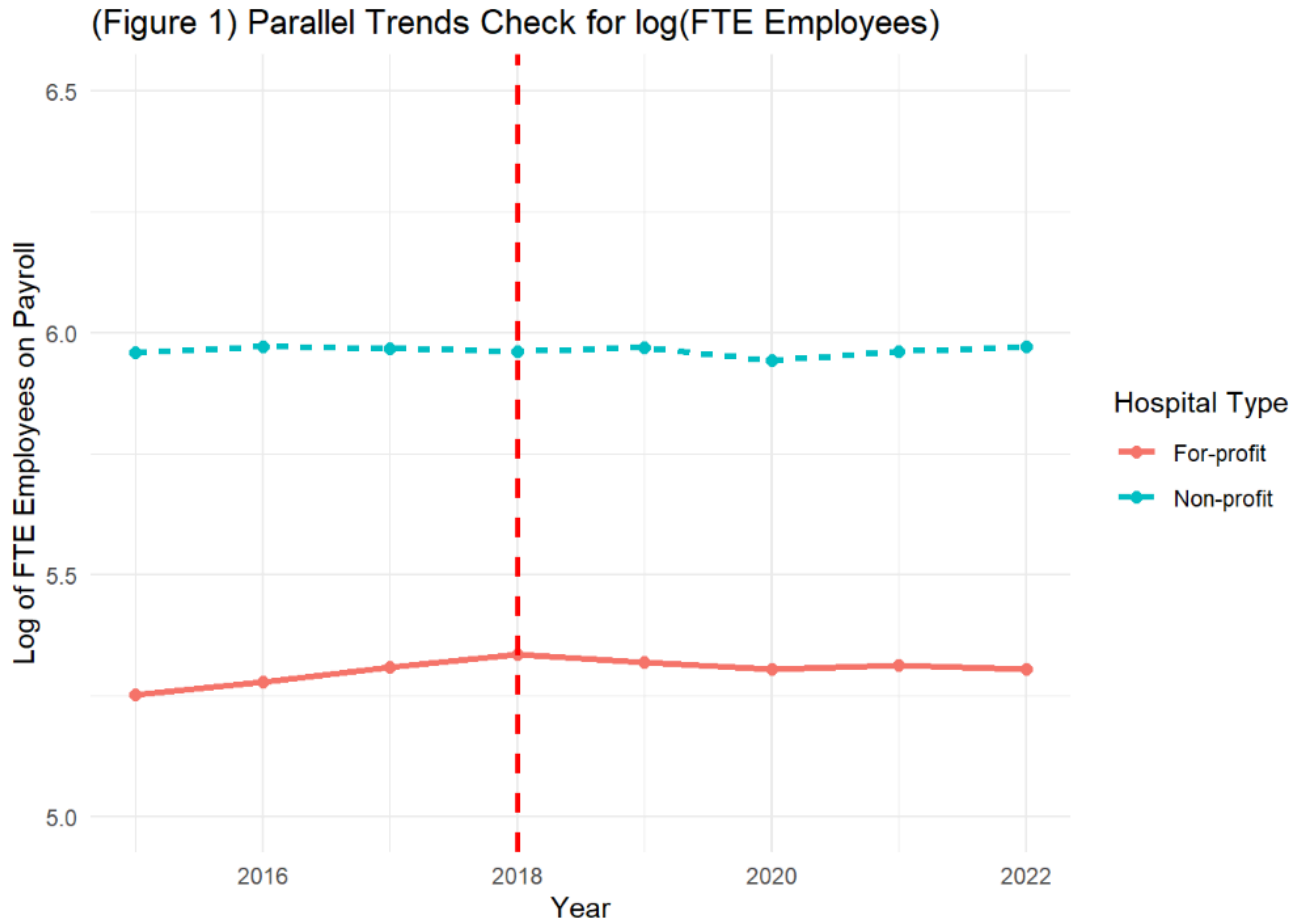


Figure 1 represents the average value of $\log(\text{FTE Employees})$ for for-profit hospitals and non-profit hospitals from 2015 to 2022. The red vertical line in 2018 represents the treatment year, as TCJA took effect on January 1, 2018. The primary focus of the parallel trend check is before the treatment, in this case, before TCJA. This plot shows a close parallel relationship between for-profit and non-profit groups before the treatment year, suggesting that the parallel trend assumption holds for the $\log(\text{FTE employees})$ variable. Furthermore, the plot shows an interesting trend for for-profit hospitals: their FTE employees decreased slightly after the TCJA act took effect, whereas non-profit hospitals remained approximately unchanged. This finding corresponds with the microeconomic theory covered in the last section.

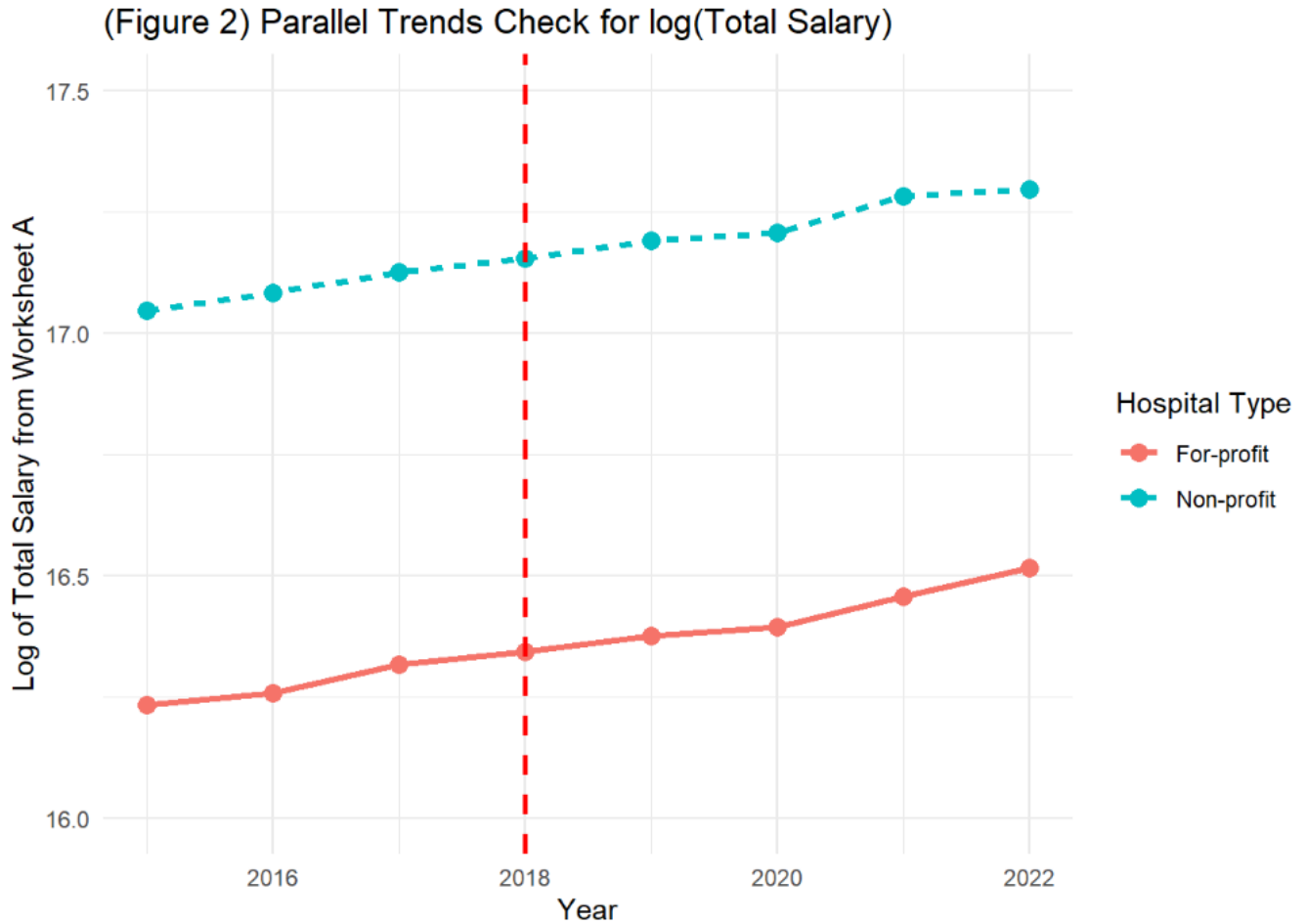


Figure 2 illustrates the parallel trend plot of log(total salaries) for non-profit and for-profit hospitals, presented in the same setting as Figure 1. There is a close parallel relationship between non-profit and for-profit entities before the TCJA in 2018. However, unlike FTE employees, total salaries remain roughly unchanged by TCJA in for-profit hospitals compared to non-profit hospitals, as the parallel relationship persists after 2018. Parallel trends hold for both primary interest variables, which enable the use of DID and first-differencing methods. The following section will cover the causal inference analysis model specification and interpret the estimation results to provide a comprehensive understanding of the actual impacts.

5. Econometrics

5.1 DID model estimation

This research estimated the simple DID model on log(FTE employees) using only the post and for-profit binary variables. The model specification is listed below:

$$\log(FTE\ Employees)_{it} = \beta_0 + \beta_1(ForProfit_i * Post_t) + \beta_2 ForProfit_i + \beta_3 Post_t + \varepsilon_{it} \quad [5.1]$$

In this specification, $\log(FTE\ Employees)_{it}$ denotes the log of full-time employees of hospital i in time period t . β_0 is the intercept, representing the average FTE employees for a non-profit hospital before the TCJA treatment. $ForProfit_i * Post_t$ is the primary explanatory variable for explaining the treatment effect. More specifically, β_1 describes the treatment effect of TCJA on FTE Employees. This coefficient is the percentage change in the number of FTE employees at for-profit hospitals after the TCJA compared to non-profit hospitals. β_2 is the baseline difference between for-profit and non-profit hospitals. β_3 represents the time effect that impacts all hospitals. Table 5 presents the estimated results from four models for 2019 to 2022, based on the dataset.

Table 5: Difference-in-Differences Regression $\log(FTE\ Employees)$ results

	DID 2017 vs 2019	DID 2017 vs 2020	DID 2017 vs 2021	DID 2017 vs 2022
Intercept	6.061*** (0.028)	6.061*** (0.028)	6.061*** (0.028)	6.061*** (0.028)
Post*For Profit	-0.004 (0.072)	-0.004 (0.072)	-0.028 (0.072)	-0.051 (0.072)
For Profit	-0.610*** (0.051)	-0.610*** (0.051)	-0.610*** (0.051)	-0.610*** (0.051)
Post	0.011 (0.039)	-0.006 (0.039)	0.000 (0.039)	0.010 (0.039)
Num.Obs.	6114	6113	6115	6105
R2	0.046	0.046	0.047	0.049
RMSE	1.28	1.28	1.28	1.28
Std.Errors	IID	IID	IID	IID

• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The primary interest coefficient β_1 in these models is close to zero and relatively consistent across years, as expected under micro theory. However, none of the coefficients are statistically significant, which may be due to omitted-variable bias or state-level economic activities that may bias the dependent variables. Interestingly, the coefficients for the post and for-profit binary variables are significant, implying that TCJA affected all hospitals similarly, as indicated by the parallel trend graph, which shows the trend persists after the treatment is administered. However, this can also be caused by the strong correlation between the binary variable and the interaction term, which increases the standard error of the interaction term and thus renders it statistically insignificant.

To combat those potential effects that cause estimations to be biased. This paper presents a model that controls for hospital size via bed numbers and total costs, hospital operational characteristics via total discharges, and location-related characteristics via an urban binary variable. In addition, it includes state fixed

effects to control for state-level economic conditions, which can influence hospitals' employment. Below is the model specification:

$$\log(FTE\ Employees)_{it} = \beta_0 + \beta_1(ForProfit_i * Post_t) + \beta_2 ForProfit_i + \beta_3 Post_t + \chi'_{it} \gamma + \lambda_s + \varepsilon_{it} \quad [5.2]$$

In this estimation specification, there are two additional elements: control variables are included in $\chi'_{it} \gamma$ while the state fixed effect is λ_s . Table 6 shows the results of this regression model. A similar pattern is evident in this table: the coefficients for the interaction term are not significant, whereas those for the binary variables are substantial, suggesting that the same problems are present in this estimation.

Table 6: Difference-in-Differences Regression log(FTE Employees) with control and fixed effects results

	DID 2017 vs 2019	DID 2017 vs 2020	DID 2017 vs 2021	DID 2017 vs 2022
Post*For Profit	0.008 (0.017)	0.012 (0.020)	-0.001 (0.019)	0.002 (0.019)
For Profit	-0.101*** (0.016)	-0.094*** (0.016)	-0.108*** (0.016)	-0.103*** (0.016)
Post	-0.051*** (0.010)	-0.083*** (0.013)	-0.135*** (0.014)	-0.183*** (0.014)
Total Discharges	0.032*** (0.006)	0.028*** (0.008)	0.026*** (0.007)	0.018* (0.007)
Total Costs	0.859*** (0.008)	0.856*** (0.010)	0.779*** (0.013)	0.792*** (0.013)
urban	-0.043*** (0.012)	-0.047*** (0.014)	-0.051*** (0.014)	-0.049*** (0.014)
Num.Obs.	7281	7280	7281	7271
R2	0.933	0.895	0.901	0.901
R2 Within	0.920	0.876	0.883	0.883
RMSE	0.34	0.43	0.42	0.42
Std.Errors	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust

• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Next, this paper employs the same method and specification to examine the impact of TCJA on the other primary interest variable: Total Salaries, with the following two equations:

1. Base DID Estimation :

$$\log(Total\ Salaries)_{it} = \beta_0 + \beta_1 ForProfit_i + \beta_2 Post_t + \beta_3 (ForProfit_i * Post_t) + \varepsilon_{it} \quad [5.3]$$

2. DID Estimation with Control and Fixed Effect :

$$\log(Total\ Salaries)_{it} = \beta_0 + \beta_1 ForProfit_i + \beta_2 Post_t + \beta_3 (ForProfit_i * Post_t) + \chi'_{it} \gamma + \lambda_s + \varepsilon_{it} \quad [5.4]$$

Table 7 presents the results for Base DID, and Table 8 presents the results for DID with control and fixed effects.

Table 7: Difference-in-Differences Regression log(total salaries) results

	DID(Log salary) 2017 vs 2019	DID(Log salary) 2017 vs 2020	DID(Log salary) 2017 vs 2021	DID(Log salary) 2017 vs 2022
Intercept	17.207*** (0.029)	17.207*** (0.029)	17.207*** (0.029)	17.207*** (0.029)
Post*For Profit	-0.004 (0.074)	-0.031 (0.074)	-0.045 (0.074)	-0.053 (0.075)
For Profit	-0.721*** (0.053)	-0.721*** (0.053)	-0.721*** (0.053)	-0.721*** (0.053)
Post	0.065 (0.040)	0.104** (0.040)	0.170*** (0.040)	0.224*** (0.041)
Num.Obs.	6114	6113	6115	6105
R2	0.059	0.062	0.065	0.067
RMSE	1.33	1.32	1.33	1.33
Std.Errors	IID	IID	IID	IID

• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 8: Difference-in-Differences Regression log(total salaries) with control and state fixed effect results

	DID(Log salary) 2017 vs 2019	DID(Log salary) 2017 vs 2020	DID(Log salary) 2017 vs 2021	DID(Log salary) 2017 vs 2022
Post*For Profit	0.010 (0.015)	0.014 (0.015)	0.013 (0.014)	0.030* (0.014)
For Profit	-0.116*** (0.014)	-0.117*** (0.014)	-0.124*** (0.014)	-0.127*** (0.014)
Post	-0.008 (0.008)	-0.017* (0.008)	-0.026** (0.009)	-0.030** (0.010)
Total Discharges	-0.003 (0.005)	-0.004 (0.005)	-0.004 (0.005)	-0.006 (0.005)
Total Costs	0.943*** (0.008)	0.944*** (0.008)	0.900*** (0.011)	0.903*** (0.010)
urban	-0.011 (0.010)	-0.012 (0.010)	-0.012 (0.010)	-0.015 (0.010)
Num.Obs.	7281	7280	7281	7271
R2	0.959	0.959	0.960	0.959
R2 Within	0.950	0.949	0.951	0.950
RMSE	0.28	0.28	0.28	0.28
Std.Errors	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust	Heteroskedasticity-robust

• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

One interesting finding among the results is that the coefficient for post * for-profit interaction term on the 2017 vs 2022 regression model is significant at the 10 % level. This coefficient implies that, holding all other variables constant, for-profit hospitals have, on average, 3% higher total salaries after the TCJA than non-profit hospitals from 2017 to 2022, implying that the TCJA's effects could take years to be fully realized. However, this finding needs to be treated with caution because it is the only significant interaction term. To rule out problems with the DID model specification, this paper implements the first-differencing model in the following subsection, using the same time period and variables.

5.2 First differencing model estimation

First Differencing (FD) involves running a regression based on the difference in the dependent variable between two time periods. FD has a more robust estimate of the causal effect because its error term structure removes time-varying effects. The specification of the FD model for log(FTE Employees) is listed below:

$$\Delta \log(FTE\ Employees)_i = \beta_0 + \beta_1 \Delta ForProfit + \Delta \chi'_i + \Delta \varepsilon_i \quad [5.5]$$

The results for this model specification are presented in Table 9.

Table 9: First Differencing Regression log(FTE Employees)

	Log (FTE Employees) 2017 vs 2019	Log (FTE Employees) 2017 vs 2020	Log (FTE Employees) 2017 vs 2021	Log (FTE Employees) 2017 vs 2022
Intercept	0.024*** (0.005)	0.041*** (0.007)	0.050*** (0.007)	0.074*** (0.008)
For Profit	-0.002 (0.008)	0.005 (0.011)	-0.013 (0.011)	-0.032** (0.012)
Number of Beds	0.181*** (0.021)	0.237*** (0.026)	0.250*** (0.023)	0.253*** (0.024)
Discharge	0.099*** (0.012)	0.156*** (0.014)	0.137*** (0.011)	0.149*** (0.011)
Num.Obs.	3057	3056	3058	3048
R2	0.053	0.082	0.101	0.112
RMSE	0.20	0.28	0.26	0.30

• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

From table 9, the coefficient for for-profit is statistically significant at 5% level in 2022. The regression model estimates that, on average, the number of FTE employees in for-profit hospitals was 3.2% lower from 2017 to 2022 compared to non-profit hospitals, controlling for hospital size. This finding further confirms the trend observed in the previous DID study and in the theoretical model.

Next, this research implements the same FD specification for $\log(\text{Total Salary})$. Below is the model specification:

$$\Delta \log(\text{Total Salary})_i = \beta_0 + \beta_1 \Delta \text{ForProfit} + \Delta \chi'_i + \Delta \varepsilon_i \quad [5.6]$$

The results for this model specification are presented in Table 10.

Table 10: First Differencing Regression $\log(\text{Total Salary})$

	Log (Total Salary) 2017 vs 2019	Log (Total Salary) 2017 vs 2020	Log (Total Salary) 2017 vs 2021	Log (Total Salary) 2017 vs 2022
(Intercept)	0.091*** (0.003)	0.150*** (0.004)	0.227*** (0.005)	0.305*** (0.006)
for_profit	-0.003 (0.006)	-0.021** (0.007)	-0.023** (0.008)	-0.027** (0.009)
d_beds	0.133*** (0.015)	0.165*** (0.016)	0.173*** (0.016)	0.198*** (0.019)
d_discharges	0.190*** (0.009)	0.155*** (0.008)	0.160*** (0.008)	0.191*** (0.009)
d_urban	-0.013 (0.015)	-0.025+ (0.013)	-0.011 (0.013)	-0.022 (0.015)
Num.Obs.	3057	3056	3058	3048
R2	0.165	0.158	0.171	0.197
RMSE	0.15	0.17	0.19	0.24

• $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The First Differencing regression estimates for $\log(\text{total salary})$ show a pattern similar to those for $\log(\text{FTE employees})$. Specifically, the coefficient terms for the for-profit binary variable are statistically significant across the three regression models for the years 2020 to 2022. Moreover, all coefficients are negative and show a decreasing trend, suggesting that, as the year progressed, for-profit hospitals had lower total salaries than non-profit hospitals, ranging from 2.1% lower in 2020 to 2.3% lower in 2021 and 2.7% lower in 2022, controlling for hospital size.

6. Conclusion

This study examined the impacts of TCJA on employment, comparing for-profit hospitals to non-profit hospitals using difference in differences (DID) and first difference (FD) methods, with FTE employees and total salaries as measures of employment and labor compensation. This study found that, despite initial stability in employment and labor compensation for for-profit hospitals, these hospitals had, on average, 3.2% lower labor force counts from 2017 to 2022. Additionally, for-profit hospitals had, on average, approximately 2.7% lower total salaries than non-profit hospitals from 2017 to 2022. Both DID and FD methods captured the causal effect of the policy intervention; these results suggest that TCJA is responsible for lower employment and lower labor compensation in for-profit hospitals. This implies that hospital ownership type influences how hospitals respond

to major tax policies, how firms' profit-maximizing strategies respond to a tax cut, and the impact of such a strategy on employment behavior. This phenomenon can be explained by the full capital expensing provision in TCJA, which made capital investment more profitable for for-profit hospitals, thereby reducing investment in employment.

One limitation of this paper is the influence of COVID, as most of the statistical significance in the models is observed after COVID. The lack of statistical significance prior to COVID-19 could pose problems for this study if COVID-19 affected for-profit and non-profit hospitals differently, thereby violating the control-group and treatment-group assumptions and rendering them incomparable. The data also face limitations, including a short time period and the potential for omitted variables that vary over time across hospitals. Future studies could extend this work by incorporating additional years of data and other employment measures to capture the timing of for-profit hospitals' responses. In addition, future studies could examine whether COVID-19 affected for-profit and nonprofit hospitals differently.

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